Power BI administration framework — options, plan and feasibility

Version 0.1 — discussion draft. Assumptions marked `[VERIFY]` must be confirmed with the customer before this becomes a commitment.

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1. What the customer actually asked for

Three questions were raised: user access management, workspace organization, and access levels inside workspaces. These are not three separate problems. They are three views of one thing: **an operating model for the platform**.

The dependency runs in one direction:

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Licensing & capacity  →  Workspace structure  →  Roles inside workspaces  →
Access provisioning
      (what you can       (where content          (what people can
(how people get
       afford / enable)    lives)                  do there)
in and out)
```

Deciding access levels before deciding workspace structure produces rework. The plan below follows the dependency order.

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2. Baseline facts you need before designing anything

Do not design in a vacuum. Phase 0 is an inventory. These are the questions that change the answer:

#	Question	Why it changes the design
1	Which licenses exist today: Pro, PPU, or a Fabric capacity (F SKU)?	F64 and above lets free-license users consume content. Below that, every viewer needs Pro. This alone reshapes the workspace and app strategy.
2	How many workspaces exist, and how many have a named owner?	Orphaned workspaces are the most common finding and the biggest cleanup cost.
3	Is Microsoft Entra ID group management delegated to IT, or can the BI team create groups?	If the BI team cannot create groups, the whole access model has to run through a ticket process.
4	Is there an existing tenant-settings configuration, and who is Fabric Administrator today?	Global Administrators acting as BI admins is a finding, not a design input.
5	Are Fabric items (lakehouses, notebooks, pipelines) in scope, or Power BI only?	Fabric item creation can be controlled separately from Power BI. Scope creep risk.
6	Is there any data classification / sensitivity labeling requirement?	Purview integration is a separate workstream if yes.
7	Are there external users (partners, contractors) consuming reports?	External sharing settings and B2B guest handling become a design constraint.
8	Is there a regulatory constraint (data residency, audit retention)?	Drives audit log export and capacity region choices.

How to collect this — three sources, in this order:

1. **Admin portal**: tenant settings screen, workspaces list, capacity settings.
2. **Fabric / Power BI REST APIs**: the admin scanner API (`getInfo`) returns a full metadata dump of workspaces, semantic models, reports, and permissions. This is the only realistic way to inventory a tenant with more than ~50 workspaces.
3. **Activity log**: 30 days of activity events give real usage — who publishes, who views, what is dead. Export to storage if you need more than 30 days.

If REST API access is not granted in week 1, the whole timeline slips. Flag this as the top dependency.

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3. Design decision A — workspace organization

The four viable patterns

Pattern	Structure	Fits when	Weakness
A. By business domain	One workspace per domain: Finance, Sales, Ops, HR	Small-to-medium orgs, single BI team, few reports	No dev/prod separation; everything is production
B. Domain + lifecycle	`Finance-DEV` / `Finance-TEST` / `Finance-PROD` per domain, wired to a deployment pipeline	Regulated or change-controlled environments	3× workspace count; needs discipline and a release process
C. Hub and spoke	Central *data* workspaces hold certified semantic models; *report* workspaces consume them via shared datasets	Multiple report authors reusing the same models	Requires Build permission governance and a certification process
D. By project or team	One workspace per initiative	Consulting / project-driven organizations	Degenerates into sprawl fastest; hardest to certify

Recommended default

C + B combined, scoped by governance tier. Concretely:

- Separate the **data layer** from the **report layer**. Semantic models live in domain data workspaces owned by the BI team. Reports live in domain report workspaces owned by the business.
- Apply **Dev/Test/Prod only to content classified as critical**. Applying it to everything is the most common cause of a governance program stalling.
- Introduce a **third tier: sandbox / self-service**, explicitly ungoverned, with a stated expiry policy. If you do not give self-service a legitimate home, it colonizes production.
- Group workspaces under **Fabric Domains** (Finance, Commercial, Supply Chain) so domain-level settings and delegated admin become possible later.

Governance tiers

Tier	Naming	Deployment pipeline	Certification	Owner
Certified / critical	`PROD-<Domain>-Data`, `PROD-<Domain>-Rpt`	Required	Required	BI team + domain steward
Managed self-service	`<Domain>-Team-<Name>`	Optional	Promoted only	Business owner
Sandbox	`SBX-<User or Team>`	No	No	Creator, 90-day review

Naming convention — pick one and enforce it via the workspace request process: `<Tier>-<Domain>-<Purpose>` → `PROD-FIN-Data`, `PROD-FIN-Rpt`, `SBX-Commercial-Pricing`.

Naming is not cosmetic. It is what makes the monthly audit script possible.

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4. Design decision B — access levels inside workspaces

Power BI workspace roles are fixed. The design work is mapping personas onto them, not inventing new ones.

Role	Can do	Assign to
Admin	Everything, including adding/removing users and deleting the workspace	2 named people maximum per workspace: BI platform team + one business backup
Member	Publish, share, add others up to Member level	Domain steward, senior analyst
Contributor	Create and edit content; cannot share externally or manage access	Report developers
Viewer	Read published content, respects row-level security	Nobody, in the recommended model — see below

The important nuance

Do not use the Viewer role as the consumption channel. Use **Power BI Apps** with audiences instead. Reasons:

- Apps give a curated experience; workspace access exposes work-in-progress content.

- App audiences let you serve different report subsets to different groups from one workspace.
- Removing someone from an app audience is a one-line change; auditing dozens of Viewer assignments is not.

Reserve the Viewer role for genuine exceptions where someone needs access to a raw item.

Row-level security interacts with roles

RLS is only enforced for Viewers and app consumers. Members and Contributors bypass it. If a domain has confidentiality requirements, the number of Contributors is a security decision, not a convenience decision.

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5. Design decision C — how access is granted

Options

Option	Mechanism	Verdict
Individual user assignment	Add people directly to workspaces	Reject. Unauditable and unmaintainable past ~20 users.
Entra ID security groups	One group per workspace-role pair, workspaces reference groups only	**Recommended.**
Entra ID + access packages (Entitlement Management)	Self-service request with approval workflow and expiry	Best end state; requires Entra ID P2 licensing `[VERIFY]`
Privileged Identity Management for admins	Just-in-time elevation to Fabric Administrator	Recommended for the tenant-admin role only

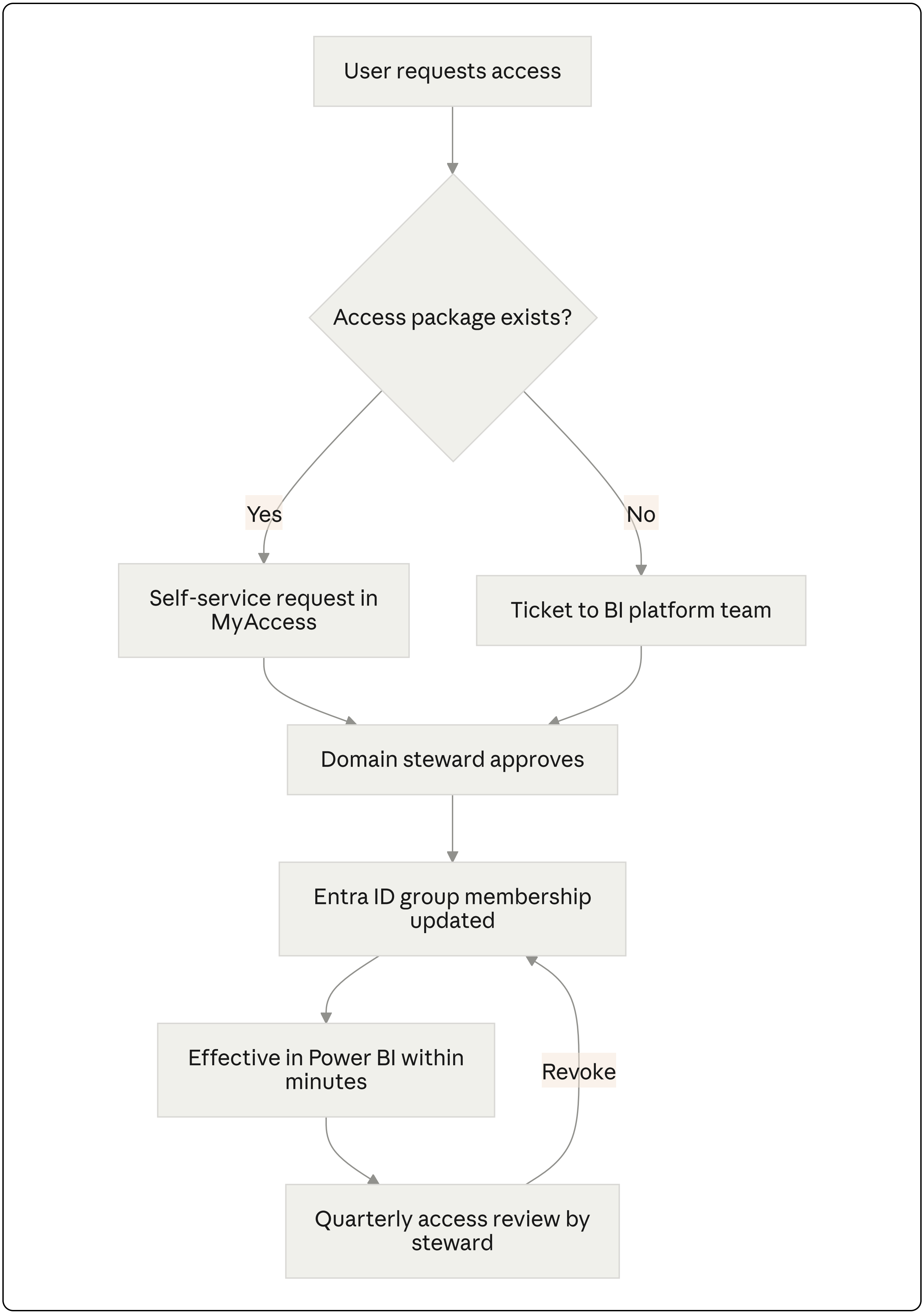
Recommended group naming

```
PBI-<Domain>-<Tier>-<Role>

PBI-FIN-PROD-Admin
PBI-FIN-PROD-Member
PBI-FIN-PROD-Contributor
PBI-FIN-APP-Consumers
```

One rule: **a workspace never contains a person, only a group.** This makes the access review a group-membership review, which HR and IT already know how to run.

Request-to-access flow



6. Design decision D — tenant settings baseline

Tenant settings are the guardrails. A minimum defensible baseline:

Setting	Recommended	Rationale
Create workspaces	Restricted to a `PBI-Workspace-Creators` group	The single highest-impact anti-sprawl control
Publish to web	Disabled for the entire organization	Creates anonymous public URLs. No legitimate internal use case.
Export data / Export to Excel	Restricted to specific groups	Bulk exfiltration path
External sharing	Restricted, guest users limited	
Publish apps to entire organization	Restricted group	
Certification	Enabled, certifier group defined	Required for the endorsement model to work
Usage metrics for content creators	Enabled	Needed for adoption KPIs
Service principals can use APIs	Enabled for a dedicated monitoring service principal	Required for automated inventory
Fabric item creation	Separately controlled `[VERIFY scope]`	Lets you govern Power BI without blocking or opening Fabric

Change control: every tenant-setting change gets a record — what changed, who approved, business justification. This is the artifact an auditor will ask for.

7. Roles and responsibilities

Role	Responsibility	Headcount
Fabric / Power BI Service Administrator	Tenant settings, capacity, gateways	2 named people (never Global Admins)
Capacity administrator	Capacity assignment, throttling monitoring	1–2
Domain steward	Certifies models, approves access, owns the semantic layer for the domain	1 per domain
Workspace owner	Accountable for content and lifecycle in one workspace	1 named person per production workspace
Content creator	Builds reports and models within the guardrails	Many
Consumer	Uses apps	Many

The role that gets skipped and shouldn't: **workspace owner**. A workspace without a named individual owner is by definition ungoverned.

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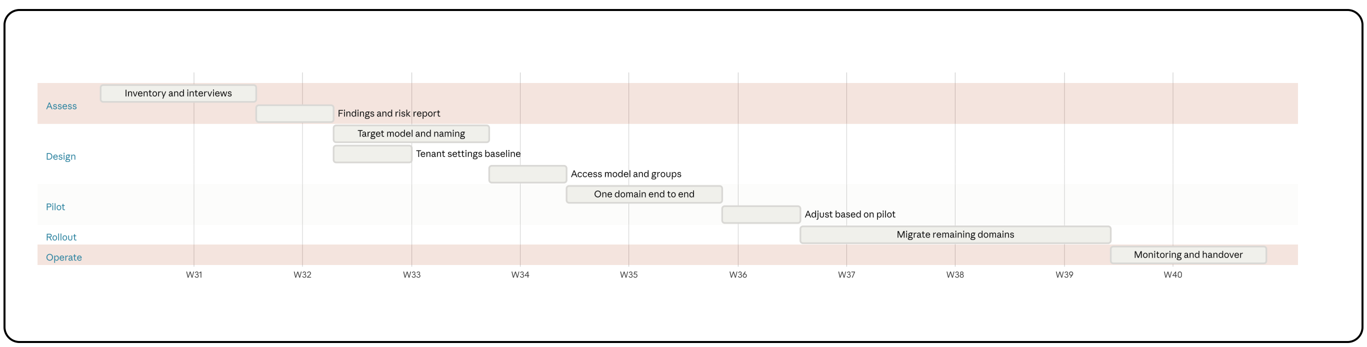
8. Execution plan



Phase	Duration	Output	Difficulty
0. Assess	2–3 weeks	Inventory, findings, risk register, current-state diagram	Low technically. Blocked by access.
1. Design	2–3 weeks	Governance framework document, workspace map, group naming standard, tenant-settings baseline, RACI	Low–medium. Mostly decisions, not build.
2. Pilot	2 weeks	One domain fully restructured and migrated	Medium
3. Rollout	4–8 weeks	All domains migrated, apps published, old workspaces archived	**Medium–high — this is where projects die**
4. Operate	Ongoing	Monitoring dashboard, quarterly access review, runbook	Low

Total: 10–16 weeks for a mid-sized tenant (roughly 50–200 workspaces, 300–1500 users). Under 30 workspaces, compress to 6–8 weeks.

Resources

Resource	Effort
BI governance lead (this role)	60–80% for phases 0–2, 40% for phase 3
Power BI / Fabric administrator	20–30% throughout — must have admin portal access
Entra ID administrator	10% — group creation, access packages
Domain stewards	4–6 hours each in design, 1 day each during migration
Executive sponsor	Decision authority at two gates: after Design, after Pilot

Estimated cost drivers to flag

- Fabric capacity SKU change, if moving to F64+ to enable free-license viewers.
- Entra ID P2, if access packages are wanted.
- No tooling purchase is strictly required. Inventory can be built with the REST APIs and a Power BI report.

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9. Feasibility and risk

Technical difficulty: low. Nothing here is hard to configure. Workspace roles, groups, tenant settings, and deployment pipelines are all native, documented features.

Organizational difficulty: high. This is the honest assessment and it should go in the customer deck. The failure modes are:

Risk	Likelihood	Impact	Mitigation
No named owners for existing workspaces	High	High	Time-boxed claim period, then archive unclaimed workspaces (with a restore window)
Users lose access to something during migration	High	Medium	Run old and new in parallel for 2 weeks; never delete, archive
Restricting workspace creation is seen as a blocker	High	Medium	Ship the sandbox tier and a fast (<48h) workspace request SLA at the same time
Admin portal access not granted in time	Medium	High	Escalate in week 1; the project cannot start without it
Scope creeps into full Fabric governance	Medium	Medium	Explicitly bound the scope to Power BI in the SOW
Licensing surprises	Medium	High	Confirm license model in phase 0, not phase 3

Viability: high, conditional on two things — a named executive sponsor, and admin-level tenant access. Without either, deliver the design document and stop; do not attempt rollout.

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10. Expected results

Concrete, measurable outcomes to commit to:

Metric	Target
Production workspaces with a named owner	100%
Workspaces created outside the request process	0 per quarter
Access granted via group rather than individual	>95%
Critical semantic models certified	100%
"Publish to web" reports in the tenant	0
Time to grant access to an existing workspace	<48 hours
Duplicate semantic models covering the same subject	Reduce by a stated % from baseline

Deliverable artifacts:

1. Current-state assessment and risk register
2. Governance framework document (this document, expanded and customer-specific)
3. Workspace map and naming standard
4. Entra ID group catalog and provisioning process
5. Tenant settings baseline with change log
6. RACI matrix
7. Migration runbook
8. Monitoring report (adoption, capacity, access, orphaned content)
9. Administrator handover and training session

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11. Open questions for the customer

1. Current license and capacity model?
2. Who has Fabric Administrator today, and will this engagement get admin access?
3. Is Fabric (beyond Power BI) in scope?
4. Are sensitivity labels / Purview required?
5. Approximate workspace, report, and user counts?
6. Is there an existing data governance body, or does one need to be created?
7. Is the outcome a design document, or design plus implementation?

Question 7 is the one to settle first — it changes the estimate by a factor of three.